

My research interests span the areas of **Software, Systems, ML, and Robotics**. In particular, I am interested in developing algorithms and tools that enable building resilient systems, deployed in dynamic environments that can automatically handle goal tradeoffs, incorporate user preferences and constraints, identify causes of failures, and self-adapt to recover from failure. I integrate *distributed systems, control theory, statistical learning and optimization, causality, and program analysis*.

1.1 Background	1
1.2 Vision: Building resilient and long-lived computer systems	1
1.3 Impact	2
1.4 Current research	3
1.5 Future Research	5

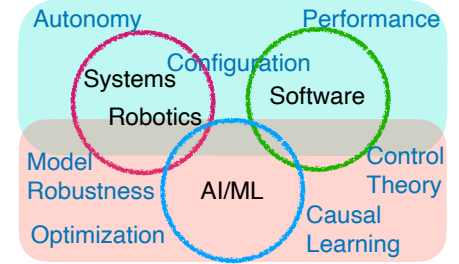


Figure 1.1: The communities that I am involved in and the topics that I work on.



Figure 1.2: As a faculty member at UofSC, I have advised 4 Ph.D. students, 3 postdocs, and worked with several graduate students and undergraduates as RA: (Top L->R) **Shahriar Iqbal** defended his Ph.D. in Fall 2023, and he is now a senior R&D engineer at Synopsys Research, working on performance optimization of distributed cloud deployment; **Fatemeh Ghofrani**, **Rasool Sharifi**, and **Abir Hossen** are 3rd year Ph.D. students; **Sonam Kharde** and **Mehdi Yaghouti** are postdocs; (Bottom L->R) **Mohammad A. Javidian** was an RA and a postdoc in my group and he is now a prof. at Appalachian; **Saeid Ghafouri** was a Ph.D. intern and is now an ML engineer at Seldon; **Kimia Noorbakhsh** was an undergraduate intern and is now a Ph.D. student at MIT; Shuge Lei, Ying Meng, and Jianhai Su were RA at AISys.

Even though I prefer the use of "we", due to tradition, here I use the term "I". My work is highly collaborative (Figure 1.5), typically across multiple institutions (e.g., CMU, Columbia, NASA). I conduct most of my research with Ph.D. students and postdocs, helping them perform research within the direction I provide.

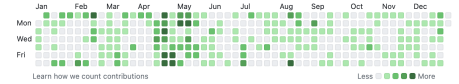


Figure 1.3: <https://github.com/pooyanjamshidi>

[1]: **Jamshidi et al.** (2017), 'Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis', *ASE*.

1: (1) not all options influence performance, (2) their influence varies, and (3) options locally interact.

[2]: **Velez et al.** (2020), 'ConfigCrusher: Towards White-Box Performance Analysis for Configurable Systems', *ASE*.

[3]: **Velez et al.** (2021), 'White-box analysis over machine learning: Modeling performance of configurable systems', *ICSE*.

[4]: **Velez et al.** (2022), 'On Debugging the Performance of Configurable Software Systems: Developer Needs and Tailored Tool Support', *ICSE*.

1.1 Background

As a professional software engineer, graduate student, and postdoc, I had the opportunity to get my hands dirty with a variety of computer systems, including infrastructure frameworks (Storm, Hadoop), large-scale cloud-native and microservice systems, ML inference pipelines (Seldon Core), robots (Husky, JPL lander), and special-purpose hardware (Jetson Xavier, Edge TPUs, FPGAs). I have explored these systems to understand their performance (throughput, power) and failure modes. For inspecting the systems, I explored a variety of interventions across the system stack, e.g., changing code, tweaking configuration knobs, and injecting disturbances. Across these experiences, several interesting themes emerged that caught my attention:

- ▶ The need to maintain goals, when systems are deployed in dynamic environments (node failure, resource variability). → My Ph.D. work
- ▶ The need to make tradeoffs between multiple competing goals (e.g., throughput and power). → My postdoc work at Imperial and CMU
- ▶ The need to identify (and reason based on) causal mechanisms impacting system performance. → The work of my team—**AISys** at UofSC

1.2 Vision: Building resilient and long-lived computer systems

I envision the deployment of resilient and long-lived computer systems that autonomously respond to changes in both the physical and computational resources available to them. To realize this vision, my research involves:

(1) *building resilient computer systems that can handle goal trade-offs*, (2) *building mechanisms for automatic diagnosis and adaptations in the event of component-level failures to avoid catastrophic system failure*, and (3) *developing verifiable algorithms to support (1)&(2) and generalize to a variety of applications*.

Resilient systems should meet user-specified goals (response time, power) while remaining responsive to dynamic changes in the environment. This is challenging for several reasons. First, these goals are *conflicting*, and customers may have *preferences* over these goals. Second, developers need *assurance* that the system remains responsive in fluctuating environments. Third, the system can be *adapted* across the system stack, from hardware to software, and the choices made between levels can *interact*.

Summary of my current research towards the vision My work addresses these challenges by enabling performance tradeoffs for computer systems that adapt to meet user-specified goals automatically and optimally. I have studied the effect of configuration interactions in highly configurable systems; specifically, I have (i) conducted **empirical studies** [1] to understand the effect of options and their interactions, the stability of the effects when the environment changes, and to provide insights¹ to motivate future research in performance analyses of highly configurable systems; (ii) developed **program analysis** techniques—ConfigCrusher [2], Compex [3]— and tools—GLIMPS [4]— to enable white-box analysis of performance behavior, by localizing *code blocks*, where options and interactions influence performance, and identifying *cause-effect chain* by identifying how influencing options are used in the implementation (see Figure 1.4). However, program analysis techniques do not scale to large systems, so I have (iii) developed **solutions** to enable performance modeling and optimization, in large-scale, multi-component, composed systems. Specifically, I have developed ML algorithms in these areas:



Figure 1.5: I value collaborations and I learned tremendously by interacting with my awesome collaborators from academia and industry.

- **Transfer learning:** To learn transferable performance models [5] and enable faster optimizations by exploiting similarities between environments [6, 7].
- **Causal reasoning:** To identify causal mechanisms, across the system stack, that affect the performance behavior of computer systems, we can employ causal reasoning to identify root causes of faults [8, 9] and these causal factors remain stable when the environment changes [10].
- **Bayesian multi-objective constrained optimization:** To enable cost-aware optimizations, when cost varies between objectives [11], and to address the difficulty in optimizing performance in locally smooth, non-convex, multi-modal behavior [12].

Finally, to provide formal assurance of behavior in dynamic environments, I have developed control theoretic solutions that are (1) robust to measurement noises or actuation delays, (2) guarantee controller stability, and (3) have bounded time to find optimal adaptation and enaction to the underlying system [13–17].

Research approach I conduct three synergistic types of research:

1. **Empirical research** to lay the foundations and enable a better understanding of a problem that we are interested in investigating. For example, I conducted a large empirical study to understand ‘why’ and ‘when’ transfer learning works for performance modeling [1]. The study revealed important insights that informed several external research² and provided a better understanding of some of the outstanding challenges in performance modeling.
2. **Practical research** by developing new methods or solutions, motivated primarily by our empirical observations, to address specific challenges in our domain of interests, in systems [8], software [3], ML [11], and robotics [9].
3. **Integrative research** by developing new theories [21, 22], infrastructure/tools/benchmarks (Figure 1.14, Figure 1.15), or data sets to allow further research³.

1.3 Impact

Industry To make an impact, I find important problems in areas of interest, then I build practical solutions to address the problems, evaluate the solutions in real-world systems, write scientific papers, publish the work in the relevant community, and disseminate the contributions through technical talks. To validate the relevance of research problems, I interact with engineers from a variety of companies to obtain their feedback on my research⁴. I believe that contributions to open-source software (Figure 1.3) can have a long-term impact, for example, through a collaboration with Intel, we integrated our cloud autoscaling controller with OpenStack [24]. Furthermore, I actively participate and contribute to relevant open-source communities⁵.

I work in the entire system stack from hardware [8] to the software level [6], and coordinate in the system stack [11]. I also work on a variety of computer systems, including stream processing systems [12], cloud [14], robotics [9], on-device AI [11], ML pipelines [25], and traditional software [3]. Two of my early work on microservices architecture migration [26] and multi-cloud migrations [27] has been used in several consulting companies as a reference methodology for migration and re-architecture projects [28–30]. These works have also been taught and used in several courses on distributed systems and cloud computing around the world⁶. I also engage in the development of community-wide guidelines for industry-relevant research [31, 32].

Academic I have received the UofSC’s 2022 Breakthrough Stars Award⁷ and the 2020 CSE Junior Researcher Award. During five years as an assistant professor at UofSC, I have advised *three postdoctoral researchers, four doctoral students, 15 undergraduates and several undergrad and graduate RA*. To support my time and the time of students, interns, and postdocs in my lab, I have collaboratively raised over **\$4 million** in competitive NSF (1 small, 1 medium->lead, 1 large, 1 EAGER), NASA (2 small->lead, 1 large->lead) and DARPA (1 sub-award) grants and around \$100K in unrestricted gifts from Google, NVIDIA, AMD/Xilinx, and Roblox. I was invited to deliver research talks at institutions

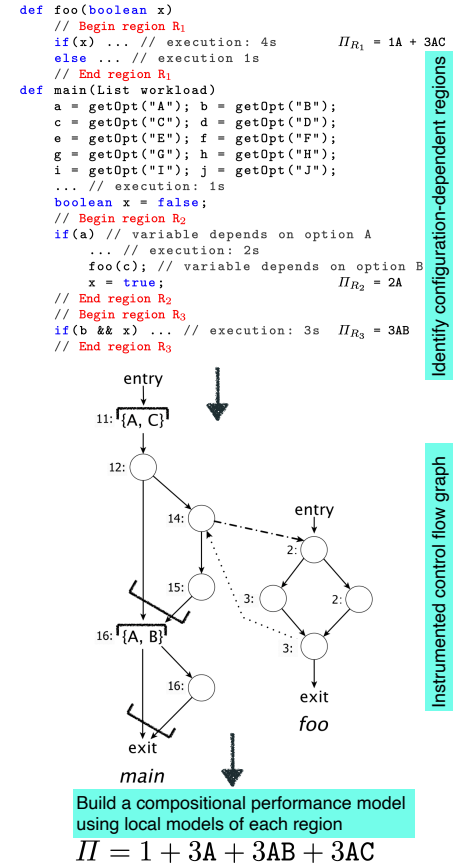


Figure 1.4: Building white-box models with program analysis: ConfigCrusher [2], Compress [3], GLIMPS [4].

[6]: Jamshidi et al. (2018), ‘Learning to Sample: Exploiting Similarities Across Environments to Learn Performance Models for Configurable Systems’, *FSE*.

[7]: Krishna et al. (2020), ‘Whence to Learn? Transferring Knowledge in Configurable Systems using BEETLE’, *TSE*.

[8]: Iqbal[★] et al. (2022), ‘Unicorn: reasoning about configurable system performance through the lens of causality’, *EuroSys*.

[9]: Hossen[★] et al. (2023), ‘CaRE: Finding Root Causes of Configuration Issues in Highly-Configurable Robots’, *RA-L*.

[10]: Iqbal[★] et al. (2023), ‘CAMEO: A Causal Transfer Learning Approach for Performance Optimization of Configurable Computer Systems’, *SoCC*.

[11]: Iqbal[★] et al. (2023), ‘FlexiBO: A Decoupled Cost-Aware Multi-Objective Optimization Approach for Deep Neural Networks’, *JAIR*.

[12]: Jamshidi et al. (2016), ‘An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems’, *MASCOTS*.

2: e.g., motivated cross-layer software variability research [18], further empirical investigations [19], taught in courses, also motivated research in other disciplines [20].

[21]: Javidian[★] et al. (2020), ‘AMP chain graphs: Minimal separators and structure learning algorithms’, *JAIR*.

[22]: Javidian[★] et al. (2020), ‘Learning LWF chain graphs: A Markov blanket discovery approach’, *UAI*.

3: Throughout my postdoc research, I created several datasets, e.g., *Storm* and *ASE17*, measuring performance metrics for highly configurable systems that allowed researchers in the community to expand their work, as reported in an independent survey [23].

such as CMU, UPenn, Google, and NASA, and invitation-only seminars, including Dagstuhl, Japan NII, Stanford MLSys, and AI Summit at Silicon Valley.

I strongly believe that active participation in research communities matters for long-term impact.⁸ I publish in AI/ML (UAI, AAMAS, AAAI, AutoML, JAIR), Computer Systems (EuroSys, SoCC, JSys, TCC, FGCS), Software Engineering (ICSE, FSE, ASE, TSE, TOSEM, TAAS) and Robotics (IROS, RA-L, T-RO) venues. I have released the code and data associated with my publications throughout my Ph.D. and postdoctoral training, and I have continued to do so as a faculty member⁹. I advocated for open-access and open-review processes in peer-reviewed venues. I cofounded JSys (Figure 1.9), a new open-access journal, covering the wide area of computer systems, and served as area co-chair in **computer architecture** (with Devashree Tripathy) and **configuration management** (with Tianyin Xu) and helped establish an efficient and transparent peer review system, making artifact evaluation mandatory for all articles. I served as PC co-chair of SEAMS 2023¹⁰, where we introduced two submission deadlines¹¹. I also served as a PC member or area chair of premier conferences (e.g., ICSE, ASE, AutoML, AISTATS), and co-organized workshops (e.g., MLArchSys at ISCA 2023). I have received ACM TOSEM distinguished reviewer award. Finally, none of these achievements would be possible without the support of my mentors (Figure 1.6).

1.4 Current research

This section is an elaboration of my current research that I summarized in Section 1.2.

Research Focus 1: ML for Systems and Systems for ML

Performance modeling and multi-objective optimization. Modern computer systems are becoming increasingly more configurable across the stack. Configurability introduces complexity: A software system with n independent binary configuration options results in 2^n different ways of configuring the system, each with different system performance. With billions of potential configurations, one cannot enumerate all configurations to find an optimal configuration. I have developed whitebox techniques (see Figure 1.4) for performance modeling [2], explainability [3], and reduce the cost of building white-box performance models [4]. I have also developed principled black-box algorithms for performance optimization, including an uncertainty-aware Bayesian optimization (BO4CO [12]) and a decoupled cost-aware multi-objective optimization (FLEXiBO [11]), and by combining offline learning and online brute search (INFADAPTER[33] and IPA [25]) and evaluated them with MapReduce-type distributed systems, on-device ML inference (see Figure 1.11), and ML inference pipelines (see Figure 1.10).

Transfer learning. Another challenge is that the relationships between system-level variables and performance metrics may dramatically change from one environment to another, so even if we build a model in one environment that relies on correlations between variables, it performs poorly in predicting performance in other environments (see Figure 1.12). I have shown that the optimal configuration in one environment is not optimal and could even perform poorly in others [6, 10, 11]. In an empirical study [1], I found that even though some relations may change, there exist invariant relationships across environments [1]. Based on this observation, I have developed L2S [6] to inform configuration sampling by learning from cheaper environments (e.g., simulators). I have also developed solutions that identify causal relationships between system variables and performance and since causal relationships are stable across environments, we were able to accelerate root causes identification of faults in ML systems (UNICORN [8]) and robotics (CARE [9]), and enable faster optimizations by learning a causal performance model from source and updating it in the target (CAMEO [10]).

Interactive optimization. Whether software or robots, these solutions have common themes. First, they multiple handle objectives and constraints (latency, power, and cost). Second, they take user-specified goals, preferences, and constraints and automatically reconfigure the system across the stack to meet those goals. This automation (1) reduces system maintenance costs and (2) allows end-user customization by automatically tailoring resource usage. However, fully automated algorithms do not have human intuitions. In a collaborative work, we investigated the effect of algorithm-human interactions to explore the architectural design space [34]. In our interactive algorithm, humans can make choices at certain intermediate points, guiding the exploration towards more desirable solutions. This resulted in the discovery of new solutions that remained unexplored in fully automated search approaches (Figure 1.13).



Figure 1.6: I owe my career to my great academic mentors: Isil Dillig (UT Austin), Ladan Tahvildari (Waterloo), Baishakhi Ray (Columbia), Christian Kaestner (CMU), Claus Pahl (Bozen-Bolzano), Homay Valafar (UofSC), Marco Valtorta (UofSC), Jason O'Kane (Texas A&M), David Garlan (CMU), and Michael Huhns (UofSC).

4: As part of our NASA-funded RASPBERRY-SI project, I have worked with two teams at NASA JPL and Ames to develop, integrate, and evaluate our autonomy solution. Also, as part of our NSF Medium collaborative project, I interacted with several startups, including Causely, DBTune, and benchANT.

5: For example, because of our recent research in AI accelerators for LLMs (see Section 1.5), I attend the OpenXLA and MLIR community meetings and contribute to the development of new features in these frameworks.



Figure 1.7: Sonam Kharde, Abir Hossen, and I visited NASA JPL in 2022 to integrate and evaluate our autonomy solution on the NASA Europa Lander testbed.

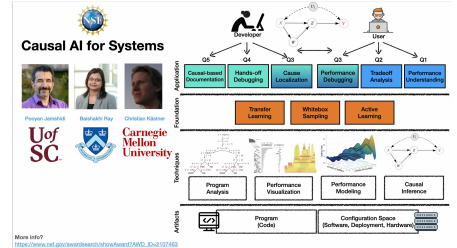


Figure 1.8: Our collaborative NSF Medium project.

6: e.g., TCSS 462/562: Cloud Computing Course at UW

7: The most prestigious university research award. Please read my interview in Breakthrough Magazine.

8: I genuinely enjoyed interacting with researchers in different communities (SE, Systems, Robotics, and AI/ML), and I learned tremendously throughout these diverse cross-community interactions.

9: AISys lab GitHub Organization

10: SEAMS is a CORE-A ranked conference on building self-adaptive and autonomous systems.

11: The introduction of two deadlines resulted in more number of papers being submitted and allowed few papers to get a revision decision instead of a reject.



Figure 1.9: with Vijay Chidambaram, Neeraja Yadwadkar, Ivo Jimenez, and Romain Jacob, we co-founded JSys, a diamond open-access journal on systems research, to lift barriers that restrict access to science.

[33]: Salmani[♦] et al. (2023), 'Reconciling High Accuracy, Cost-Efficiency, and Low Latency of Inference Serving Systems', EuroMLSys.

[25]: Ghafouri[★] et al. (2023), 'IPA: Inference Pipeline Adaptation to Achieve High Accuracy and Cost-Efficiency', JSys.

Research Focus 2: Autonomous Systems and Robotics

Robust control of uncertainty in dynamic environments. I have spent significant effort, during my Ph.D., creating an infrastructure for studying and benchmarking self-adaptive systems, focusing in particular on handling different sources of uncertainty in dynamic environments such as cloud [35]. Specifically, I have developed ELASTICBENCH and integrated it with Microsoft Azure and OpenStack (see Figure 1.15). This work raises several challenges, including conflicting user preferences, uncertain delays in cloud resource provisioning, and measurement noise. My research has addressed these challenges by (i) developing techniques for robust estimations and propagation of uncertainty, (ii) developing control theoretic solutions with quantifiable robustness to error, and (iii) applying the solutions in a variety of computer systems—cloud applications, mobile robots, and robotic platforms for space exploration.

Robust uncertainty estimation. Performance engineers often have to estimate the performance of a software system before having full knowledge of the system parameters, such as workload and operational profile. These uncertain parameters inevitably affect the accuracy of quality evaluations and the ability to judge if the system can continue to fulfill performance requirements if parameter results are different from expected. Previous work has addressed this problem by modeling the potential values of uncertain parameters as probability distribution functions and estimating the robustness of the system using Monte Carlo-based methods that require a large number of samples, resulting in high computational costs and long waiting times. I addressed this issue by extending Polynomial Chaos Expansion for uncertainty propagation, demonstrating orders of magnitude faster analysis and accurate estimations of uncertainties in software systems [36, 37].

Robust control theory. When designing a self-adaptive system, we would like guarantees that the system will adapt to uncertainties and fluctuations in the deployed environments. The main challenge here is to build a controller that is robust against different sources of uncertainty to avoid oscillations in resource allocation due to temporary small variations in the monitored values. To address this challenge, I have designed several robust cloud auto-scaling controllers: (1) ROBUST2SCALE [13]—can handle conflicting adaptation policies, (2) FQLAKE [14]—learns optimal adaptation policies without user input, and (3) HMC [15]—a synthesized PID controller for memory management in computer systems. In collaborative work, we have designed a reference control design process that automatically synthesizes control systems for self-adaptive software systems [38]. The controllers produced by this technique guarantee that they will converge to the desired goal, provide bounds on the convergence time, and have quantifiable robustness to error. Later, to bridge the gap between software engineering and control theory disciplines, we summarized the result of many years of research in the control field and discussed open challenges, where I primarily contributed to mechanisms for dealing with uncertainty in control strategies for adaptation [17]. I extended these techniques to control multiple goals simultaneously while maintaining formal guarantees [15]. A novel contribution of this work is a generalization of classic control techniques so that the controller can be developed without specific knowledge of the system under control.

Autonomy for space explorations. I am also excited about autonomy in space explorations, primarily because of the unique autonomy challenges in this area. I am leading a multi-institution NASA-funded project, RASPBERRY-SI (Figure 1.16), in collaboration with CMU, York, UArk, NASA JPL, and NASA Ames. Space robotic exploration has many unique challenges: extremely low temperatures, high radiation levels, lengthy communication delays, limited bandwidth, and the difficulty of maintaining communication links. Spacecraft power will be restricted to onboard systems (batteries, etc.), which limits avionics performance and places a premium on efficient use and management of power. To address these challenges, we have developed a solution¹² based on causal reasoning that identifies the root causes of unpredictable faults due to operation in an unknown and uncertain environment and autonomously adapts the activities of the spacecraft to recover from failure [9]. We also showed that the causal mechanisms are stable from simulation to physical robots, so it allows faster and more reliable debugging (CURE [39]; see Figure 1.17 and Figure 1.18). I have also contributed to the development of a reusable infrastructure¹³ with modular building blocks to support the instantiation of autonomy for different target systems. We have used the infrastructure in two robotics projects—BRASS MARS¹⁴ (mobile robots) and RASPBERRY-SI (space lander).

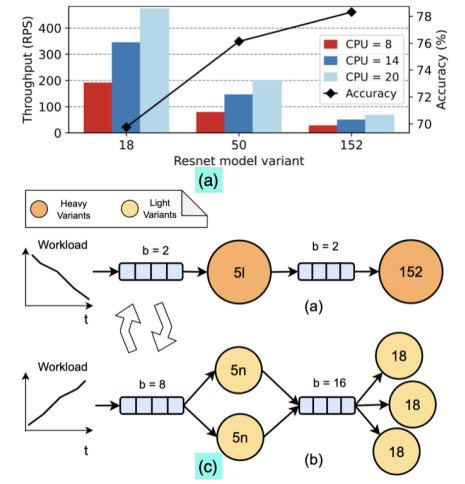


Figure 1.10: (a) INFADAPTER [33] combines model-switching and auto-scaling to tradeoff accuracy, cost, and latency in inference serving systems. (b) IPA [25] dynamically re-configures ML inference pipelines (model-switching, auto-scaling, batching) to optimize accuracy, minimize costs, and meet SLAs.

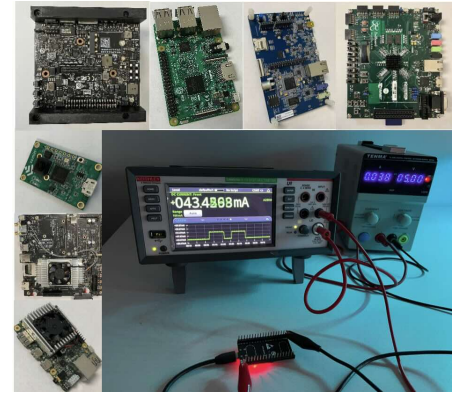


Figure 1.11: Hardware platforms in our lab students and I use for software-hardware co-design experiments.

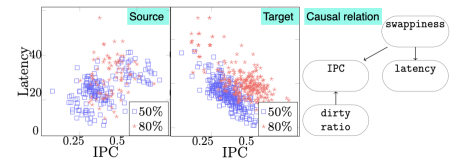


Figure 1.12: The relationship between interprocess communication (IPC) and latency reverses from source to target while the relationship between swappiness and latency stays invariant. The learned causal model also identifies that swappiness is causally influencing latency, while the behavior of IPC may change based on swappiness and dirty ratio (please see CAMEO [10]).

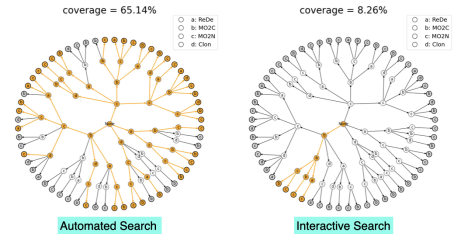


Figure 1.13: Algorithm-human interactions found novel solutions compared to fully automated search [34].

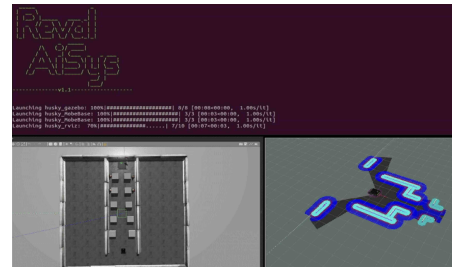


Figure 1.14: We developed Reval as a benchmark framework to evaluate the performance of robotics platforms.

Research Focus 3: Scalable Causal Transfer Learning and Structure Learning (Theory)

Transfer learning, the ability to learn a task in one environment and transfer it to another environment to learn faster with less cost, is at the heart of my research endeavor. One of the most critical problems in transfer learning is the domain adaptation task to make accurate predictions when the distributions for the source and the target differ [41]. I looked at domain adaptation from a causal perspective: learning a causal model that describes the data-generating mechanisms, which are invariant across the different domains, allows one to transfer knowledge from one domain to the other to address the problem of domain adaptation. The main idea is to select the subset of characteristics $\mathcal{A}$ that leads to the best predictions of Y in the source domains while satisfying the invariance $P(Y|\mathcal{A})$ across all domains. Unfortunately, existing approaches do not scale beyond dozens of variables due to exponentially large conditional independence (CI) tests. To overcome the problem of scalability, we proposed Scalable Causal Transfer Learning (SCTL) [42] that takes advantage of local computation by finding the Markov blanket of the target variable, drastically reducing the search space. As a result, CI tests become more reliable in high-dimensional, low-sample size scenarios. I have also developed scalable causal structure learning algorithms [43] and offered sound algorithms for learning chain graphs in high-dimensional spaces [21, 22, 44]¹⁵.

Research Focus 4: Adversarial Machine Learning

Building a generic framework that allows a tradeoff between robustness and cost. I am excited about this new line of research on the security and robustness of ML systems. Most existing work has focused on designing defense mechanisms with specific assumptions, which typically do not allow for trade-offs between robustness and cost. One way to achieve such a tradeoff is to design the system by allowing model ensembling. Although previous work showed the vulnerability of ensemble models, we showed that it is possible to construct effective ensembles by ensembling *diverse* weak defenses (WD) that complement each other. Based on this insight, we built ATHENA (see Figure 1.19), a generic framework that allows the tradeoff by adding WDs. I and my student worked with Google to implement this framework in production.

Robust representation learning. I am also enthusiastic about foundational work to gain a better understanding of model robustness in deep representation learning. In a recent work [45], we posed questions to investigate the effects of adversarial training on self-supervised and supervised contrastive learning alongside supervised learning:

- ▶ *Is there anything special about the robustness of the contrastive learning representation?*
- ▶ *To what extent does employing the label benefit or deteriorate the robustness?*
- ▶ *How does adversarial training affect the learned representations?*

Our analysis uncovers significant disparities between adversarial and clean representations in standard-trained models. Specifically, contrastive learning without label information is less robust compared to others. However, combining the standard contrastive loss with either supervised cross-entropy or supervised contrastive loss can improve the robustness of the learned representations by leveraging the label information. Our study reveals important insights regarding the impact of adversarial training on representations and can enable the designing of more effective robust models.

1.5 Future Research

The current transition to a post-Moore era is an opportunity to look well beyond power efficiency to make carbon and other sustainability metrics first-order concerns in computing. I plan to build on my current work on cross-stack performance tradeoffs to gear my future work towards enabling **sustainable computing** with a focus on software-hardware co-design, by defining projects on sustainability-aware software/system abstractions, hardware/software cross-layer co-design, and compiler optimizations with sustainability metrics¹⁶. I target LLM inference pipelines for my immediate future work, mainly due to my familiarity with LLM pipelines [10, 25] and unique systems challenges due to large model sizes and complex computations in inference.

1. Automatic hardware-aware partitioning and mapping for multi-chiplet AI inference

The execution of LLM models often exceeds the capability of single or multi-chip solutions as these models either have extensive parameters or operate on large batch size, sequence, and hidden length, demanding substantial memory and bandwidth for

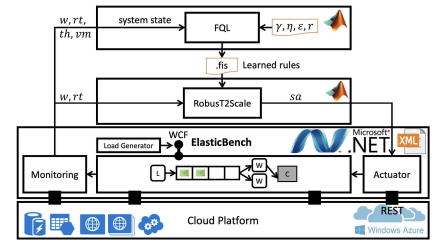


Figure 1.15: I have developed **ELASTICBENCH** [14], a cloud application framework that allows users to plug in auto-scaling logic and experimentally evaluate controllers in a feedback control loop. I evaluated **ROBUST2SCALE** [13] and **FQL4KE** [14] on **ELASTICBENCH** during my Ph.D. research.

[38]: Filieri et al. (2015), ‘Software engineering meets control theory’, **SEAMS**.

[17]: Filieri et al. (2017), ‘Control Strategies for Self-Adaptive Software Systems’, **TAAAS**.



Figure 1.16: I am leading **RASPBERRY-SI**, a large-scale multi-institution project funded by NASA.

12: including infrastructure for integrating our autonomy with NASA virtual and physical testbeds: <https://github.com/nasa-raspberry-si/>

[39]: Hossen[★] et al. (2023), ‘CURE: Simulation-Augmented Auto-Tuning in Robotics’, **T-RO**.

13: <https://github.com/cmu-mars>

14: The reusable infrastructure played a pivotal role in a third-party evaluation by MIT Lincoln Lab [40].

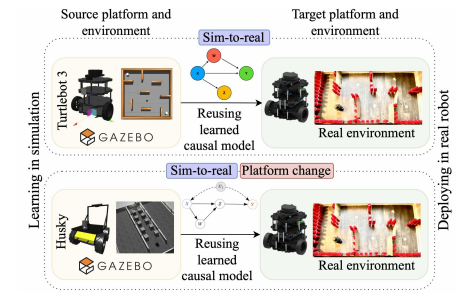


Figure 1.17: CARE [9]: identifies root causes of failure using causal reasoning; CURE [39] enables sim-to-real by learning transferable causal mechanisms.

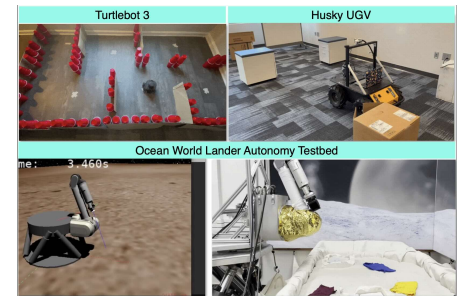


Figure 1.18: Robotic experiments my students and I designed in my lab as well as NASA JPL. We also released robot experiment videos at **CaRE**, **CURE**, and **Reval**.

[41]: Javidian[▲] et al. (2021), ‘Scalable Causal Domain Adaptation’, **WHY-21 @ NeurIPS**.

[42]: Javidian[▲] et al. (2023), ‘Scalable Causal Transfer Learning’, **JAIR**.

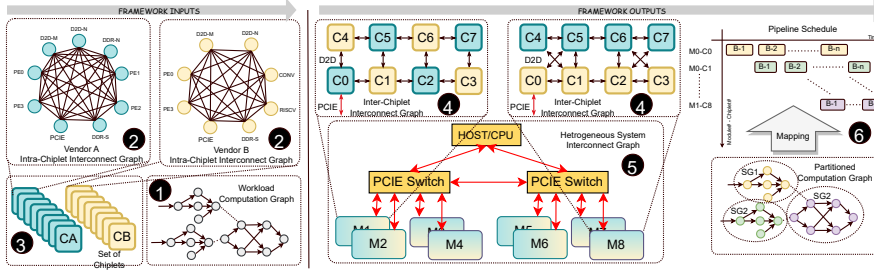


Figure 1.21: Designing multi-chiplet modules for LLMs by optimal model partitioning and mapping.

weights and reusable intermediate tensors, e.g., caching Keys/Values in transformers, to prevent unnecessary calculations. The inadequate on-chip capacity is compensated by refreshing the weights/activations from off-chip memory or spreading parameters across more chips. The problem is optimal *Partitioning* and *Mapping* of the LLM workload to the multi-chiplet hardware accelerator to maximize #Inference/Joule.

I plan to implement a framework for finding the near-optimal model parallelization plan (tensor and pipeline parallelism) of LLMs for multi-chiplet systems. Machine learning tasks constructed in PyTorch, TensorFlow, or JAX are translated into a high-level computational graph like StableHLO. This created graph undergoes target-independent graph optimizations within OpenXLA. This optimized graph is taken by our framework and transformed into a data-centric graph, introducing an additional layer of abstraction to represent the data transfer between operations. Utilizing this data-centric graph along with the high-level system network graph which models the inter-/intra-connects and on-/off-chip memory subsystem, we deploy various search techniques to find a near-optimal parallelization plan for the designated ML workload. In the end, the high-level data-focused graph is annotated with the candidate parallelization strategy before being handed over to the compiler for performance and sustainability evaluation. The calculated cost is then used to guide the search.

2. Optimizing LLM inference by finding a tradeoff between caching and recompute

Central to LLMs is an autoregressive transformer architecture that produces text token-by-token based on the provided prompt and previously generated tokens. This sequential token generation entails evaluating the model repetitively per request until an end-of-sequence marker is produced. Such a serial approach results in compute underutilization during inference, limiting the serving throughput. One approach to improving the throughput is to cache prior dynamic states, thereby avoiding redundant computations. In the case of Transformer architectures, the Key and Value tensors of the attention mechanism comprise these reusable states. With the cached attention Keys and Values, referred to as the KV cache, we can reduce subsequent computational requirements at the expense of greater memory utilization and increased data movement.

While KV caching helps mitigate redundant computations during autoregressive decoding, efficient management of this cache in real-time remains challenging under the stringent memory constraints of AI accelerators and bandwidth limitations of chip interconnects. Due to the autoregressive nature of token generation, it is impossible to predict the output sequence length of an input prompt. Therefore, if a request takes longer than expected, the inference-serving system preempts the request and swaps its KV cache to secondary memory storage (i.e., host) to free the dynamic memory pool for other requests. When the scheduler decides to continue token generation for the preempted request, it can choose whether to fetch the request KV cache from the host memory or discard it and recompute the KV tensors.

The key idea here is rather than resorting to the extremes of either swapping or recomputation for the Key-Value (KV) tensors associated with sequences, I plan to adopt a middle-ground solution. In practice, if a sequence has generated a substantial number of tokens before preemption, such as 200 tokens, the approach involves fetching the KV tensors for the initial portion (e.g., the first 100 tokens) from memory while simultaneously recomputing the KV tensors for the subsequent portion (e.g., the remaining 100 tokens). By doing so, it effectively capitalizes on concurrent computation and memory access, mitigating latency, and enhancing the overall efficiency of vLLM processing. The aim is to design a holistic serving system balancing efficient resource usage against prompt response times for real-time interactive applications. Our proposed approach introduces a balanced strategy that optimizes the overlap of computation with memory, thereby reducing latency.

15: <https://majavid.github.io/structurelearning/>

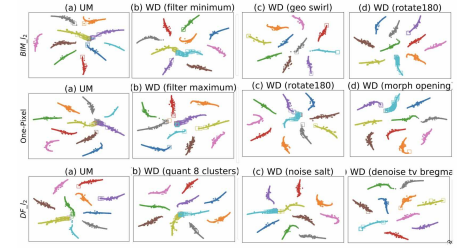


Figure 1.19: ATHENA was built based on two insights: the effectiveness of weak defenses varies; weak defenses complement each other. So, the more diverse the weak defenses, the more robust the defense becomes.

[45]: Ghofrani[★] et al. (2023), ‘Rethinking Robust Contrastive Learning from the Adversarial Perspective’, *AdvML @ ICML*.

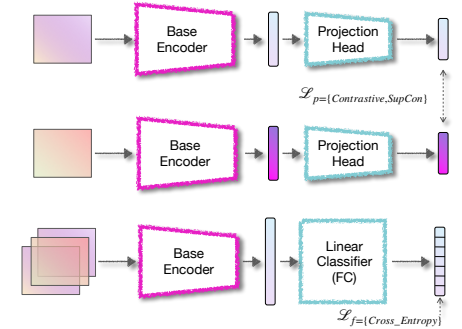


Figure 1.20: Comparing the robustness of learned representations in contrastive, supervised contrastive, and supervised learning.

16: In an NSF project, in collaboration with CMU and RIT, we investigate the broader role of software-driven sustainability. We investigate how software can enable sustainable modular system design but also influence the consumer’s behavior. One of the tasks in the project is to define metrics that enable the characterization of sustainability aspects in software design.

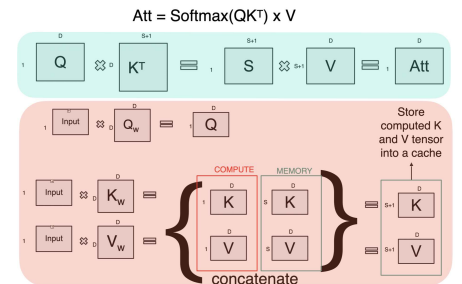


Figure 1.22: Finding a tradeoff between caching and recompute in LLM inference.

My philosophy of teaching centers on motivating students to acquire the foundations and skills required for learning, not only teaching the course materials but also showing them how they can generalize to new situations. More importantly, due to new technologies such as ChatGPT, it becomes apparent that memorizing knowledge is becoming outdated, so I teach students how to ask the right questions and the reasoning skills, by creating an atmosphere where all students are put into a position to succeed.

2.1 Teaching Experience

I have accumulated a breadth and depth of lab and classroom teaching experiences. As an undergraduate, I served as a lab assistant for the required core C++ programming course that was offered for several majors, including engineering, sciences, and computing at Amirkabir University of Technology. After receiving my Master's degree, I became a lecturer at Kharazmi University and taught software engineering courses for several semesters. During my Ph.D. at DCU, I co-lectured cloud architecture alongside my advisor. During my postdocs, I guest lectured in an ML course at Imperial College and a software design course at CMU. While at UofSC, I have been teaching three courses:

- ▶ **CSCE 585: Machine Learning Systems:** A mixed graduate and undergraduate course that I have designed from scratch to enable students to design modern ML systems and learn how to deal with unique challenges that only occur when building AI/ML systems in production. Specifically, I have designed the course suitable for connecting theory with practice, so students gain practical skills that companies like Facebook and Google are looking for.
- ▶ **CSCE 212: Introduction to Computer Architecture:** This is a required core course that provides a first introduction to computer architecture. I completely restructured the course by covering two types of instruction set architectures, MIPS and LC-3, so students can learn about the foundation of computer architecture by contrast and design decisions in each architecture. In addition, I teach the broader view of computer architecture, from problem specification to algorithms, high-level code, assembly, and machine code. I teach them about design decisions by discussing the pros and cons of each design, specifically their impact on performance. I show how performance trade-offs are made by designing simple systems and discussing decisions along the way. I design assignments to come up with alternative designs so that they become familiar with the system design process and learn the skills to become an architect.
- ▶ **CSCE 580: Artificial Intelligence:** I restructured the class to cover modern topics in AI, including deep neural networks and causality. I designed several interrelated assignments with an integrated auto-grading. In these interrelated projects, students can design real-world AI agents that cover the concepts I teach in the course, so students can write code and see how the agent behaves and how the theory that they learn in class can turn into a real working system, and with the auto-grading, students can get instant feedback.

2.2 Teaching Evaluation

I strongly believe in eliciting student feedback. I incorporate their suggestions, and I get motivation based on their positive comments to continue improving my teaching practices. Across the 4 courses I have taught in the past 5 years, my course evaluations scored a median of 4.7/5.0 and I, as the instructor, scored a median of 4.8/5.0, and once a 5.0/5.0, in most cases they were well above department and college average. The peer, chair, and unit evaluations have ranked me from Excellent to Outstanding. I received compliments in anonymous student evaluations^{1,2,3,4} and peer evaluations⁵.

2.3 Teaching Philosophy and Implementation Strategies

I have always followed a few basic principles in my teaching: preparation of high-quality content, engaging every student in the class, delivering the simplest possible way as if I am also learning alongside students, listening to students and getting feedback from peers, reflecting on my practice, and continuously learn about teaching methods.

2.1 Teaching Experience	7
2.2 Teaching Evaluation	7
2.3 Teaching Philosophy and Implementation Strategies	7
2.4 Curriculum Development	8

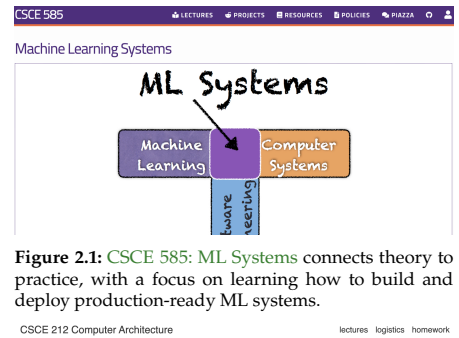


Figure 2.1: CSCE 585: ML Systems connects theory to practice, with a focus on learning how to build and deploy production-ready ML systems.



Figure 2.2: CSCE 212: Computer Architecture covers the technical foundations of how a computing platform is designed from the bottom up.

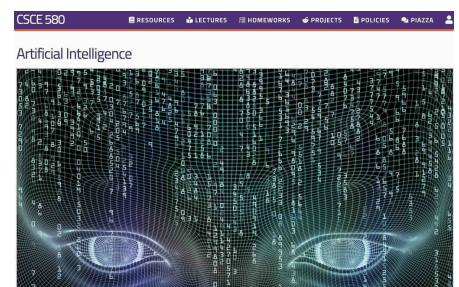


Figure 2.3: CSCE 580: Artificial Intelligence introduces the basic ideas and techniques underlying the design of intelligent computer systems.

1: "Dr. Jamshidi is amazing. I love how genuinely excited he gets about the subject matter. . ."

2: "The professor, Dr. Jamshidi, was very fair in his grading and very flexible about how to operate the class. When he noticed that the students responded well to one teaching format that he used, he decided to incorporate that format into more lectures. He cared about making sure the students actually learned the material and was enthusiastic about the course content."

3: "Dr. Jamshidi has always been encouraging and caring with what he says and all his given feedback. I appreciate the respect and kindness that many professors lack."

4: "Dr. Jamshidi was an amazing teacher and educator. I understand that his primary responsibility is as a researcher, but I cannot commend him enough for his ability to teach students the material. I was not expecting the material to be so engaging and for Dr. Jamshidi to support his students the way that he did. I had heard that he has strong requirements for his students and this is true, but it is all in pursuit of our education."

5: "Prof. Jamshidi was extremely well-prepared for the lecture. His presentation was clear and authoritative. He appeared very knowledgeable and confident. More importantly, there was particularly good interaction between students and instructors. Students came up with really good questions without being prodded. I found myself caught up in the excitement and asked a few questions myself. Of course, it is easier to have good interaction in a special topics course with a small number of students. Nonetheless, the dynamic was impressive. Prof. Jamshidi displayed pedagogical skills that are surprising for a first-semester Assistant Professor."

Teaching strategy: undergraduate vs. graduate My approach in undergraduate and graduate courses is different. In *undergraduate courses*, rather than mechanical memorization of concepts, I emphasize understanding simple concepts and connecting them with the concepts that they have learned before, so students can utilize them later in other courses. I believe that understanding a concept is more valuable than memorizing a specific solution. In addition, I try to be more hands-on and prepare tutorials with code to create an interactive learning environment. In *graduate courses*, I emphasize on bigger picture of the concepts and composition of them to formulate good questions. I discuss recent advances and open challenges and highlight possible directions. This way, I teach the state of the art and recent results but also show them how they could have come up with those publishable ideas.

Learning by analogy and discussing alternative solutions I believe that there are always alternative ways to solve a problem, perhaps each with different pros and cons. I typically discuss alternative solutions for specific problems in class and ask students to discuss the pros and cons. When students implement their own solutions they can learn and generalize better by discussing alternative solutions implemented by their peers, as they are exposed to a variety of more or less optimal implementations. I also believe that we as teachers, whenever possible, illustrate concepts with concrete examples or analogies to the previously learned concepts. These illustrative examples should be diverse to help students construct mental models that generalize well to new situations.

Feedback and improvements In addition to getting feedback from students, I also ask my colleagues to attend my lectures and give me feedback for improving my teaching skills, either in the form of informal feedback or as the requirements for yearly academic evaluations. As a member of our teaching community, I frequently discuss teaching methods with colleagues in the CSE department and beyond especially members of our broader community. I learn from others' experiences, how they utilize available tools, and refining methods for integrating our research with teaching content. To improve my teaching, I read literature in learning sciences and education [46–49].

Learning sciences and education I believe that experience alone is not sufficient to deliver a good lecture, and that even the best teachers need to continuously improve their practices. Therefore, I have taken pedagogy classes⁶ that introduced me to cognitive load theory, active learning, and self-explanation. I implemented these techniques by (1) preparing simple slides with a short text, and figures and, when required some annotated equations, effectively using multiple modalities for learning so students can learn and remember the materials with less information overload; (2) creating hands-on activities, primarily based on python notebooks, that contain the right amount of text, equation, code (see Figure 2.5); (3) exposing students to new concepts from different perspectives, including intuitive definition, to the underlying math, and annotated code to correspond the concept with implementation; (4) providing the opportunity for students to also teach to their peers (see Figure 2.6).

Inclusive teaching I am committed to creating an environment where irrespective of their background, all students feel equally valued. Not only I am committed to the ideals of inclusive teaching, but I have also incorporated them into my teaching. For instance, in the ML systems course, I invited two research scientists at Google and a female researcher at Roblox to deliver technical talks. My goal was to present the students with a variety of possible career options.

Industry-relevant teaching When I was in school, I could connect and learn better when my teachers explained why I needed to learn new concepts or techniques and in what scenarios I may need it in the future. I also found out that when I contextualize the content, students engage better and can recall the material when I ask them in the following lectures. My vision is to help students by giving them a foundation of success for them to build upon. I also make the course materials relevant to the industry. For example, I received unsolicited praises from practitioners, who watched the ML Systems lectures online.^{7 8}

2.4 Curriculum Development

I am one of the PIs of our [RTG: Mathematical Foundation of Data Science at the University of South Carolina](#), an NSF-funded program with a budget of \$2M. We developed a curriculum for a joint undergraduate program between CS, Math, and Statistics departments that covers the mathematics of ML, practical ML, and dealing with large and small data. We also train undergraduates and postdocs in collaborative research projects to advance toward a future career in academia or industry.

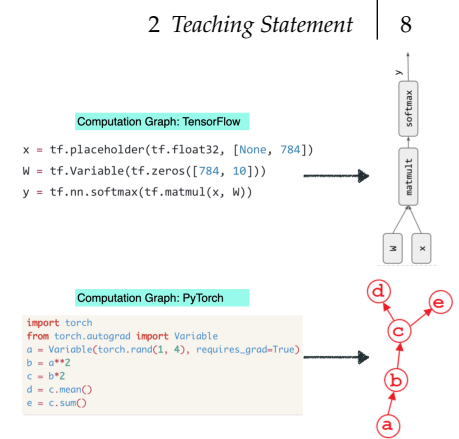


Figure 2.4: Making an analogy between computational graph in TF and Pytorch in my ML Systems class.

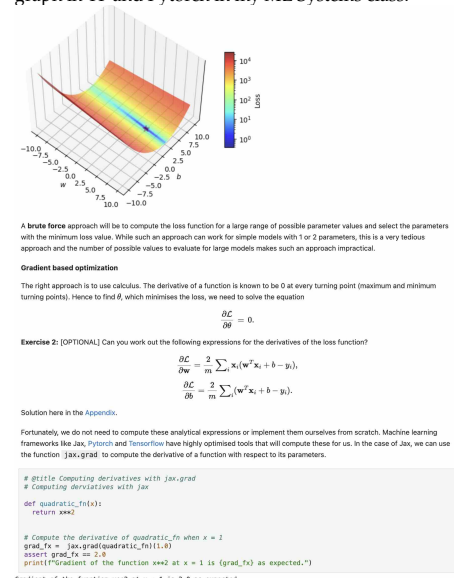


Figure 2.5: A tutorial in my ML Systems class. I typically combine multiple modalities for better learning, in particular, for teaching the mathematics of ML, I found out that if I connect equations with code snippets and visualizations, students engage better, ask more questions, and remember the concepts better.



Figure 2.6: In a recent intervention in my computer architecture course, I asked each student to teach something new about computer architecture that is tightly related to the concepts that they have learned in the class. This intervention was eye-opening for me in terms of student's enthusiasm for preparing high-quality materials and how other students engaged in the class when their peers were presenting.

6: I have also participated in several UofSC Center for Teaching Excellence workshops, including *Maintaining Effective Communication*, *Entering Mentoring*, *Writing Your Teaching Philosophy*, *Co-Teaching*. I gave an invited talk at the Oktoberfest UofSC's annual teaching symposium, and I shared my experience in the course, entitled *Exponential Learning by Building Real-World AI Systems*.

7: "Dear Pooyan, I enjoyed reviewing the material of your ML Systems course, thanks! I have to admit such content at the intersection of production and research is quite scarce and your course enabled me to build ML Systems in a principled way." Joachim Hagege, Deep Learning Engineer @ Alibaba Israel

8: "Hi Prof. Pooyan Thanks for connecting. I enjoyed your publicly available course CSCE 585, and helps me to connect the Theory/Science aspect of my work to what I am doing here at Ford, building ML Platform! Best, Anupam" Anupam Banerjee, Platform Product Manager, Machine Learning Platform

For the first couple of students that I advised, I frankly faced many challenges primarily due to some wrong assumptions. One of those incorrect assumptions was that I thought everybody learns the same way as I do, which was obviously incorrect, each mentee has a different background and upbringing, their approach to learning is different from each other, and they were exposed to a different level of breadth and depth of their program. Therefore, I had to revisit my understanding of student mentorship. My vision in mentoring students is to give them a foundation of research and problem-solving to be able to become independent researchers or more capable practitioners that distinguish them from the crowd.

During my career, I have mentored many students, including high school students, undergraduates, graduates, and postdocs. I always find it very rewarding to help others succeed. The opportunity to help students understand the subject matter, encourage them to think deeply about the core insights, and, most importantly, excite them to learn and grow is my main passion for choosing an academic career over industry.

3.1 Mentoring Philosophy

I have always aspired to achieve the same level of personal connection with my students and have worked hard to design my courses to motivate students to learn. I have always followed a few basic principles in my mentoring: getting to know students by having frequent and regular interactions and devising a personalized mentoring plan that works for that particular student.

Inclusive and personalized research mentoring In my humble opinion, the key to improving this situation is by actively encouraging minority students to engage in research early on¹, and highlighting potential benefits of a research experience beyond academia². I believe that understanding the differences between students with different backgrounds and tailoring individual research plans is crucial to successful mentoring. In the past, I have had opportunities to work or collaborate with students from multiple countries, including many European countries, Nigeria, Pakistan, and Malaysia and most of them were female. These experiences exposed me to the challenges of mentoring students from different educational backgrounds. For example, I realized that computing curriculums are vastly different between countries. Some curricula emphasize more theoretical aspects, while others place greater weight on course projects. Many minority students have a limited opportunity to learn about research because they attend undergraduate institutions where research activity is limited. The isolation that many minorities experience in school also limits their chances of learning [50]. Based on these experiences, I actively expose and recruit underrepresented students to research, either via course projects in my machine learning systems class or by recruiting them as undergraduate researchers during the semester.

Building a diverse research lab Despite the high pressure of publishing in top venues and receiving prestigious grants, I have always aspired to create a diverse lab, where I welcome people of any race, religion, national origin, gender identity, family commitment, political affiliation, sexual orientation, and eligible age or ability. As a junior faculty member, recruiting competitive graduate students is a key to success. However, I made it explicit to myself to put diversity as important and even more important than the competitiveness of the group that I am building. Toward that vision, I hired female postdocs, female Ph.D. students, and female undergraduates, and was able to build the most diverse lab in our department in terms of the number of female students.

Building a diverse robotics team One strategy that I believe will make a long impact is to bring students from diverse socio-economic levels and racial categories to solve difficult problems together, building technical skills and empathy at the same time. For example, I am the academic mentor for the university robotics team, called Gamecock Robotics, where I am so proud that we were able to build a team with a majority of female undergraduate students. To build such a diverse team, I worked with university initiatives for recruiting underrepresented minorities and women, such as Women in Computing, Minority in Computing, and McNair Scholar.

3.1 Mentoring Philosophy 9

3.2 Mentoring Strategy 10

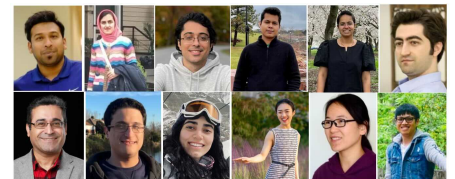


Figure 3.1: Artificial Intelligence and Systems Laboratory (AISys).

1: e.g., by expanding undergraduate research programs

2: acquiring skills such as independent thinking and problem-solving, which are also highly valued in the industry



Figure 3.2: I serve as the academic mentor for the Gamecock robotics team.



Figure 3.3: AISys social activities.

Maintaining a high standard while considering student's well-being I hold a high level of standard for the work that comes out of my group. This strategy resulted in publications well above the department's average and students who were proud of their achievements. The strategy also provided members with amazing opportunities that are distinguished from the opportunities that other students in our department can get. For example, one of my graduate students landed a research internship at Google, which never happened for any student in our department. At the same time, I am fully aware that graduate school is difficult, regardless of one's identity or background. Therefore, I have equipped myself with supportive strategies to avoid psychological traps, especially for female students and visible minorities. I have been paying attention to the student's well-being by recognizing their personal needs and the challenges they may deal with. I provide support to those who have faced such challenges by giving them more flexibility, referring them to university health and well-being services, organizing social activities with my students, and showing them that I care for them.

3.2 Mentoring Strategy

Undergraduates While at UofSC, I have mentored over a handful of undergraduate students. Undergraduate students require more hands-on mentoring support. I typically match them with one of my graduate students or postdocs and ask them to have short daily meetings, so they can resolve any blockers. I also meet with them every week. This provided them the opportunities for internships, admissions to top universities, research careers in industry, and the pleasure of discovering new knowledge or gaining insights. I have heard from some mentees, several years after their graduation, how the research experience in my lab helped them in their career³. I received funding from NSF and UofSC to support students. My mentees have gone on to Google, Microsoft, Boeing, national labs, and graduate programs at MIT, Stanford, UC Irvine, and Purdue.

Graduates I have advised four Ph.D. students and worked with several graduate-level RAs. My graduate students have been my top priority and I hold a high level of standard for their work. This strategy resulted in fewer overall publications but with quality well above the department's average and students who were proud of their achievements⁴. I meet with students at least every week and discuss results and possible directions. For early-stage students, I show them how to find interesting ideas and formulate good questions. Later on, I discuss why students ran this or that experiment, what the results mean, and whether the results can provide a viable answer to the research questions. I found out that learning how to design good experiments is the most fundamental skill graduate students need to learn. So, for their first project, I walk them through designing experiments that are aligned with the questions. I experienced that even for their second papers, students may have trouble in designing their experiments, but they typically fully learn it for their third work and they can independently design their experiments⁵. I also run a weekly **ML system reading group** (Figure 3.5) to catch up with state-the-art, but more importantly, to discuss the writing style, their design of experiments, and alternative solutions and possible ideas to improve the work.

Postdoctoral training I mentored three postdocs. Having done two postdoctoral trainings myself, I am fully familiar with the ups and downs of this temporary position. My mentoring plan for each postdoc is individualized based on his/her previous background, research experiences, and academic interests. I work with postdocs intensely at the beginning to find the right topic for their research that is connected to their previous expertise but also allows them to explore new directions. I also expose them to possible collaborations and provide co-mentoring and career advancement opportunities. For example, one of my earlier postdocs, **Mohammad Ali Javidian** had a background in mathematics and his interests were in theory and causality. He co-advised students with me and published outstanding papers in venues such as JAIR and UAI and collaborated with undergraduates, Ph.D. students, and other collaborators of mine. I also invited Mohammad Ali for guest lectures in my AI and ML Systems courses. He is now an assistant professor at Appalachian State University. Sonam Kharde, another postdoc in my group, with a Ph.D. in control theory, joined my group in early 2022. By exposing her to interesting problems and involving her in our collaborative NASA project, she was able to learn causal inference with probabilistic model checking and publish outstanding contributions. She has now gained the confidence to become an independent researcher.



Figure 3.4: I always try to find a way to make my interactions with students and mentees to be fun and pleasant.

3: Dear Professor Jamshidi, It's been a while since we last interacted at UofSC, and I fondly recall the invaluable experience I gained under your mentorship during my sophomore and junior years. The time spent in your research lab working on neural network compression (using Pruning, Knowledge Distillation, and Quantization-Aware Training) and robustness significantly influenced my career path. Since graduating, I've joined Boeing Research & Technology as an AI/ML Technologist, dedicating my efforts to Machine Learning applications in autonomous systems. Within the DARPA Assured Autonomy program, my role primarily revolves around the verification and validation (V&V) of learning-enabled components (LEC) in safety-critical autonomous systems. This endeavor has been rewarding, culminating in a publication with Imperial College London here and an upcoming publication with UC Berkeley.

4: even though after facing a few rejections, some of them complained that other graduate students have published many papers so far, I convinced them to keep up the good work and they do not need to worry about the numbers, instead focusing on the quality work and taking care of their mental health, and eventually, once they were able to get their papers accepted at top notch venues and how those few papers helped them to land good opportunities, they appreciated my encouragements and support.

5: At this stage I treat them as my peers, so they can gain confidence and move towards becoming an independent researcher in the field.

AlSys Reading Group

Figure 3.5: ML systems reading group.

Since starting my career at UofSC, I have been active in promoting diversity and broadening the participation of underrepresented groups in our department, motivated by a personal story that I have shared publicly¹. In summary, I experienced a shocking gap between the diversity that I expected from my work environment and the actual diversity in the classes that I taught. In the classes that I taught at UofSC, the number of students from underrepresented groups has been very low (around 10%). Since I have seen much better diverse environments at CMU during my postdoctoral studies, I have decided on several targeted activities to broaden the participation of females and blacks in computing, mainly focusing on the CSE department at UofSC, but also several activities in the state of South Carolina.

I believe that establishing an inclusive and supportive environment for students of all genders, ethnicities, cultures, and educational backgrounds is key to improving diversity in computer science. As a faculty member, I have actively contributed to four directions in the goal of creating inclusive environments: (1) establishing an inclusive research group and encouraging students from diverse backgrounds to participate in research, (2) organizing community-building activities to diversify research in technical areas, (3) using best practices that I have learned from validated pedagogical research in teaching, and (4) participating in outreach activities at schools.

Inclusive research mentoring In my humble opinion, the key to improving this situation is by actively encouraging minority students to engage in research early on², and highlighting potential benefits of a research experience beyond academia³. I believe that understanding the differences between students with different backgrounds and tailoring individual research plans is crucial to successful mentoring. In the past, I have had opportunities to work or collaborate with students from multiple countries, including many European countries, Nigeria, Pakistan, and Malaysia and most of them were female. These experiences exposed me to the challenges of mentoring students from different educational backgrounds. For example, I realized that computing curriculums are vastly different between countries. Some curricula emphasize more theoretical aspects, while others place greater weight on course projects. Many minority students have a limited opportunity to learn about research because they attend undergraduate institutions where research activity is limited. The isolation that many minorities experience in school also limits their chances of learning [50]. Based on these experiences, I actively expose and recruit underrepresented students to research, either via course projects in my machine learning systems class or by recruiting them as undergraduate researchers during the semester.

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1: <https://pooyanjamshidi.github.io/diversity/>



Figure 4.1: Artificial Intelligence and Systems Laboratory (AISys).

2: e.g., by expanding undergraduate research programs

3: acquiring skills such as independent thinking and problem-solving, which are also highly valued in the industry



Figure 4.2: AISys social activities.

provide support to those who have faced such challenges by giving them more flexibility, referring them to university health and well-being services, organizing social activities with my students, and showing them that I care for them.

REU site in robotics Since I started my career at UofSC, I was invited by the PI of the REU site in robotics in our department, Jason O’Kane, to serve as a mentor for some of the REU students over the summers. The overall goal of the site is to broaden participation in computing by introducing a diverse group of students to research in the discipline. I actively supported the local organizing team of the REU robotics program by helping with recruitment, review, reporting, and seminar talks, in addition to student mentoring. In addition to the site activities, I have received REU supplements associated with my active NSF grants and trained several undergraduate students by involving them in our research projects and a few resulted in high-quality publications that I indicated in my bibliography.

Creating diverse communities As a member of the scientific community, I get frustrated by the fact that the diversity level of technical events is quite low, and based on my observations, these events are typically male-dominated. For example, when I first participated in a Dagstuhl seminar, only one female researcher attended the event. I believe that as a member of the research community, we have a responsibility to increase the diversity in our scientific events. To improve this situation, when I co-organized a Dagstuhl seminar, we took three important actions: (i) we increased female participation by sending invitations to several female researchers whom we knew in our research community; (ii) we also decided to make a public call for participation, something that is rare for invitation-only events; (iii) we also asked some senior faculties to encourage their female students in their group to apply for our Dagstuhl seminar. Our main intention was to find researchers from unrepresentative cultures and backgrounds.⁴

While at UofSC, I continued the same care. One strategy that I believe will make a long impact is to bring students from diverse socio-economic levels and racial categories to solve difficult problems together, building technical skills and empathy at the same time. For example, I am the academic mentor for the university robotics team, called Gamecock Robotics, where I am so proud that we were able to build a team with a majority of female undergraduate students. To build such a diverse team, I worked with university initiatives for recruiting underrepresented minorities and women, such as Women in Computing (Yan Tong), Minority in Computing (Csilla Farkas), Multicultural Outreach Student Team (MOST), and McNair Scholar.

Inclusive teaching My view on diversity is derived from my early teaching experiences as a junior lecturer at Kharazmi University, Iran. After an initial struggle to convince students to engage better, I found that the main reason was that the groups were made up of separate genders⁵. I then decided to reform the groups by mixing them. After this intervention, the productivity of the groups increased considerably as a result of better collaboration between students.⁶ Teaching several courses at UofSC⁷ taught me that inclusive teaching can come down to subtle details. For example, while all students are capable of learning the material, they do not all arrive with the same level of academic preparation. There can be differences in the confidence and tendency of students to raise their hands that are associated with both gender and national identity or their upbringing. In terms of teaching, I apply best practices about how to avoid implicit bias in the classroom [51]. For instance, I try to allow all students to participate, speak, and ask questions in class. I also create a diverse team of TAs for my courses since they are direct role models for the students. As a teacher, I consciously called for a balance of girls and boys⁸ so that no one would come away with the impression that their teacher was biased against them.

Outreach activities One of the activities that helped me to improve my mentoring skills, especially for diverse backgrounds was engaging with primary students and non-technical people to describe my research work. I delivered talks at Claflin⁹ and other HBCUs in South Carolina, as well as primary schools in Columbia, SC¹⁰. Together with my collaborator, Karina Liles, I have participated in the Live Connect STEM Truck¹¹. STEM Truck is a mobile learning space that provides STEM activities to students and educators. These outreach and dissemination activities will always be a part of my teaching and research activities.



Figure 4.3: Serving as a mentor in the REU in Robotics program at UofSC.



Figure 4.4: A Dagstuhl seminar on performance and DevOps that I co-organized.

4: While individual events may not have an immediate visible impact, I believe that a sustained, long-term effort in this direction will be crucial to building a diverse community.



Figure 4.5: I serve as the academic mentor for the Gamecock robotics team.

5: somehow a cultural habit in Iran

6: I also noticed a higher engagement of students; they asked more and better questions and volunteered to present their progress and discuss the issues that they faced during the coursework.

7: and mentoring several students from different backgrounds, countries, and genders.

8: even though there are only a few female students



Figure 4.6: Discussing my career path with students in a local elementary school.

9: the most recent one was a virtual talk on August 28, 2020, where I talked about our robotics research to sophomore and junior seminar class

10: the most recent one was early 2020 in Windsor Elementary school that has over 90% black students

11: <http://www.stemtruck.org/>

The following are my overall representative publications that I am truly proud of. The first two represent my recent work as a faculty member at UofSC, while the last two represent the work I have done during my postdoctoral training at Imperial College London and CMU. I also provided the links to the open-access versions of the papers.

1. (2023) **FlexiBO**: A cost-aware multi-objective Bayesian optimization algorithm. ¹
2. (2022) **Unicorn**: A causal reasoning approach for performance debugging of configurable systems. ²
3. (2019) **MARS**: an integrated learning and quantitative planning approach that enables runtime reconfiguration of autonomous robots. ³
4. (2016) **BO4CO**: A method for multi-objective optimization of stream processing systems. ⁴

1: Md Shahriar Iqbal, Jianhai Su, Lars Kotthoff, and **Pooyan Jamshidi**. 'FlexiBO: A Decoupled Cost-Aware Multi-Objective Optimization Approach for Deep Neural Networks'. In: Journal of Artificial Intelligence Research (**JAIR**). 2023

2: Md Shahriar Iqbal, Rahul Krishna, Mohammad Ali Javidian, Baishakhi Ray, and **Pooyan Jamshidi**. 'Unicorn: Reasoning about Configurable System Performance through the Lens of Causality'. In: European Conference on Computer Systems (**EuroSys**). 2022

3: **Pooyan Jamshidi**, Javier Cámara, Bradley Schmerl, Christian Kaestner, and David Garlan. 'Machine Learning Meets Quantitative Planning: Enabling Self-Adaptation in Autonomous Robots'. In: 14th IEEE/ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**). 2019

4: **Pooyan Jamshidi** and Giuliano Casale. 'An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems'. In: Modeling, Analysis and Sim. of Computer and Telecomm. Systems (**MASCOTS**). 2016

Here is a list of representative publications based on the communities that I am involved in.

Computer Systems:

1. Md Shahriar Iqbal, Ziyuan Zhong, Iftikhar Ahmad, Baishakhi Ray, and **Pooyan Jamshidi**. 'CAMEO: A Causal Transfer Learning Approach for Performance Optimization of Configurable Computer Systems'. In: ACM Symposium on Cloud Computing (**SoCC**). 2023
2. Md Shahriar Iqbal, Rahul Krishna, Mohammad Ali Javidian, Baishakhi Ray, and **Pooyan Jamshidi**. 'Unicorn: Reasoning about Configurable System Performance through the Lens of Causality'. In: European Conference on Computer Systems (**EuroSys**). 2022
3. Saeid Ghafouri, Kamran Razavi, Mehran Salmani, Alireza Sanaee, Tania Lorido-Botran, Lin Wang, Joseph Doyle, and **Pooyan Jamshidi**. 'IPA: Inference Pipeline Adaptation to Achieve High Accuracy and Cost-Efficiency'. In: Journal of Systems Research (**JSys**). 2023
4. **Pooyan Jamshidi** and Giuliano Casale. 'An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems'. In: Modeling, Analysis and Sim. of Computer and Telecomm. Systems (**MASCOTS**). 2016

Software Engineering:

1. **Pooyan Jamshidi**, Norbert Siegmund, Miguel Velez, Christian Kaestner, Akshay Patel, and Yuvraj Agarwal. 'Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis'. In: Int'l Conf. Automated Software Engineering (**ASE**). 2017
2. Miguel Velez, **Pooyan Jamshidi**, Norbert Siegmund, Sven Apel, and Christian Kaestner. 'White-box analysis over machine learning: Modeling performance of configurable systems'. In: Proc. Int'l Conf. on Software Engineering (**ICSE**). 2021

Robotics:

1. Md Abir Hossen, Sonam Kharade, Bradley Schmerl, Javier Cámara, Jason M O'Kane, Ellen C Czaplinski, Katherine A Dzurilla, David Garlan, and **Pooyan Jamshidi**. 'CaRE: Finding Root Causes of Configuration Issues in Highly-Configurable Robots'. In: IEEE Robotics and Automation Letters (**RA-L** and **IROS**). 2023
2. Md Abir Hossen, Sonam Kharade, Bradley Schmerl, Jason M O'Kane, David Garlan, and **Pooyan Jamshidi**. 'CURE: Simulation-Augmented Auto-Tuning in Robotics'. In: IEEE Transactions on Robotics (**T-RO**). 2023
3. **Pooyan Jamshidi**, Javier Cámara, Bradley Schmerl, Christian Kaestner, and David Garlan. 'Machine Learning Meets Quantitative Planning: Enabling Self-Adaptation in Autonomous Robots'. In: 14th IEEE/ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**). 2019
4. **Pooyan Jamshidi**, Miguel Velez, Christian Kaestner, Norbert Siegmund, and Prasad Kawthekar. 'Transfer Learning for Improving Model Predictions in Highly Configurable Software'. In: Software Engineering for Adaptive and Self-Managing Systems (**SEAMS**). 2017

AI/ML:

1. Md Shahriar Iqbal, Jianhai Su, Lars Kotthoff, and **Pooyan Jamshidi**. 'FlexiBO: A Decoupled Cost-Aware Multi-Objective Optimization Approach for Deep Neural Networks'. In: Journal of Artificial Intelligence Research (**JAIR**). 2023
2. Mohammad Ali Javidian, Marco Valtorta, and **Pooyan Jamshidi**. 'Learning LWF Chain Graphs: A Markov Blanket Discovery Approach'. In: Conference on Uncertainty in Artificial Intelligence (**UAI**). PMLR. 2020
3. Mohammad Ali Javidian, Marco Valtorta, and **Pooyan Jamshidi**. 'AMP Chain Graphs: Minimal Separators and Structure Learning Algorithms'. In: Journal of Artificial Intelligence Research (**JAIR**). 2020

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I indicated graduate students★, undergraduate students♦, and postdocs▲ from my team as well as collaborators from industry■. Other co-authors are my academic collaborators.

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